Beyond Static Benchmarks: A Post-Leaderboard Evaluation Paradigm for Generative Music and Creative AI

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Evaluation Crisis in Generative Music

The current state of evaluation in generative music is methodologically fragmented, relies on ambiguous validation objectives, and builds on systematically misaligned metrics [1]. This has led to a crisis in the field, where the following failures are observed:

- **Validity failure:**
  - Utilized metrics are *unreliable proxies* and fail to measure musical qualities.
- **Leaderboard failure:**
  - Single-score optimization leads to *overfitting* and is *insufficient for assessing a multi-dimensional cultural artifact*.
- **Comparability failure:**
  - *Different metrics, feature representations, and reference data* prevent meaningful comparisons across models.
- **Reproducibility failure:**
  - *Closed-source representations and evaluation scripts* and proprietary data hinder reproducibility.
- **Governance failure:**
  - *Ad-hoc, fragmented evaluation protocols* indicate lack of standardization.

Goals of the Proposed Framework

- **Community-driven ecosystem governance:**
  - coordinated by the community for the community.
  - metrics and evaluation protocols have to *evolve with the field*.
- **Multi-dimensional evaluation:**
  - acknowledging the *multi-dimensional nature of music*.
- **Interpretability:**
  - utilizing *interpretable metrics* and features where possible.
- **Domain adaptability:**
  - acknowledging *variety of use cases* through curated reference benchmark profiles (RBPs).
- **Modularity & extensibility:**
  - modular software architecture allows for easy *integration of new features and evaluation methods*.
- **Reproducibility & transparency:**
  - versioned *open source* software with comprehensive documentation and *auditable logging*.

Governance Structure

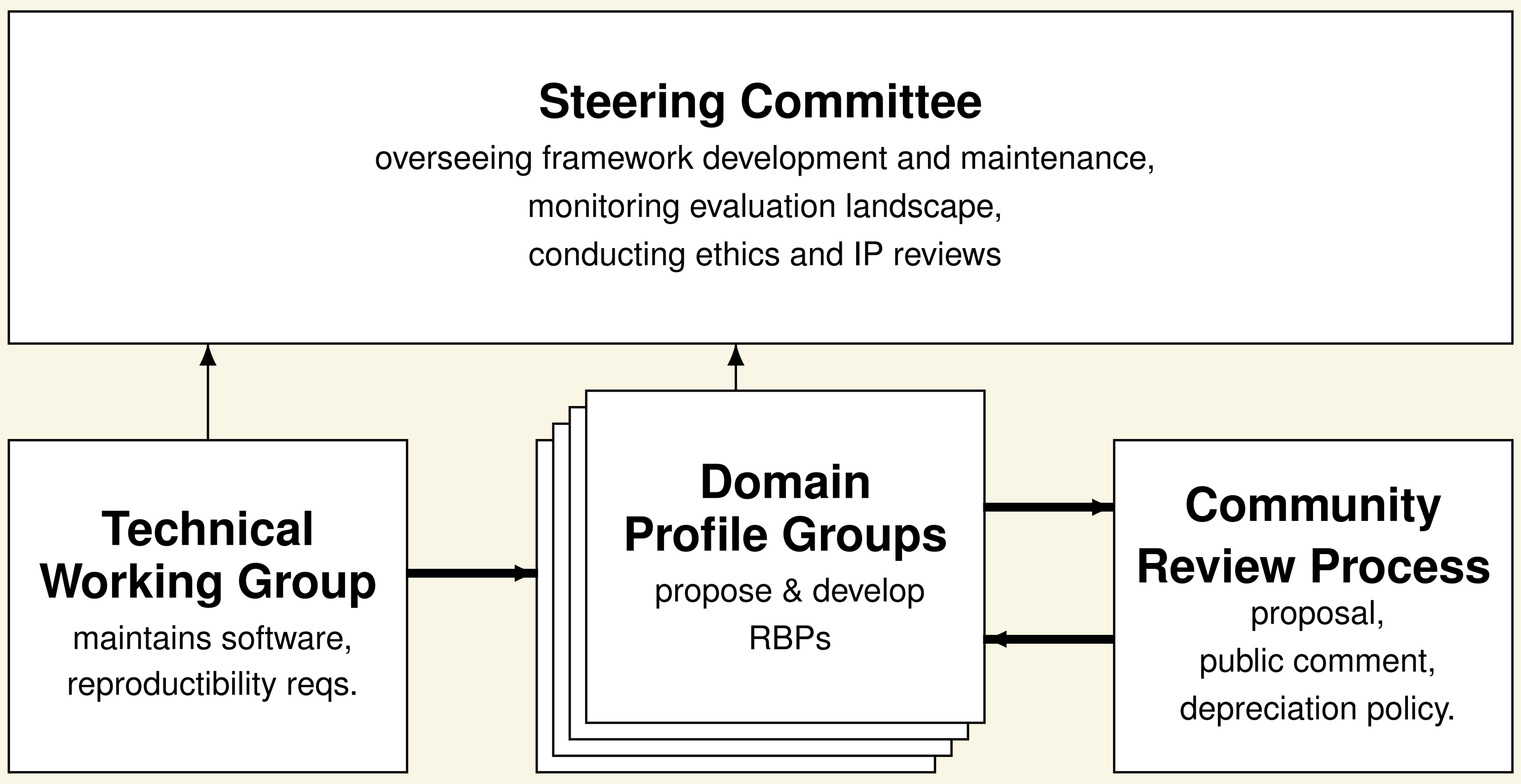

Figure: Proposed governance structure

Architecture

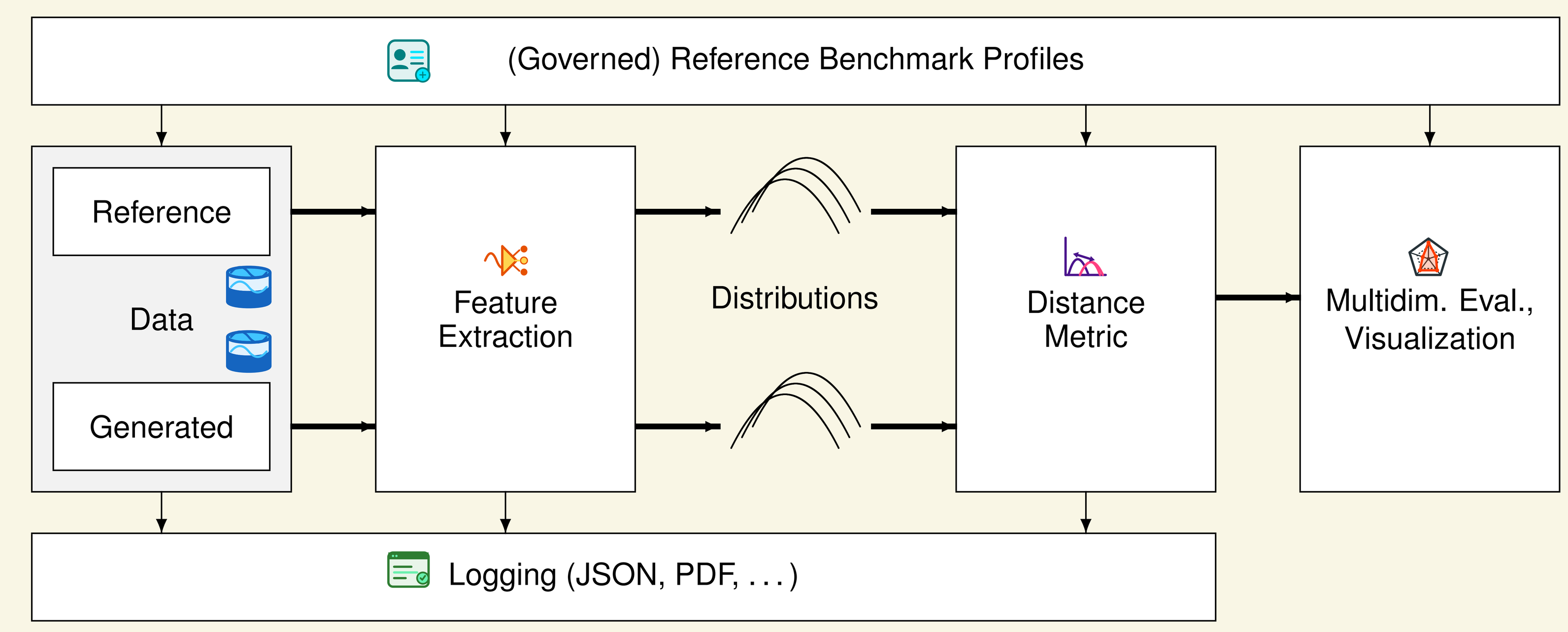

Figure: Proposed framework architecture for the evaluation of generative music.

- **Reference Benchmark Profiles (RBPs):**
  - use-case specific combination of *reference data*, *feature representations*, and *distance metrics*, and *reporting format*.
- **Standardized Reference Data:**
  - publicly available, documented datasets as controllable open references.
- **Standardized Feature Representations:**
  - musical qualities
  - audio fidelity
  - controllability
  - originality
- **Standardized Distance Metrics:**
  - distributional distance metrics (Wasserstein, KL, MMD, etc.)
  - traditional evaluation metrics (accuracy, precision, recall, F1, etc.) for classification tasks
  - task-specific metrics as needed
- **Multi-dimensional Results & Visualization:**

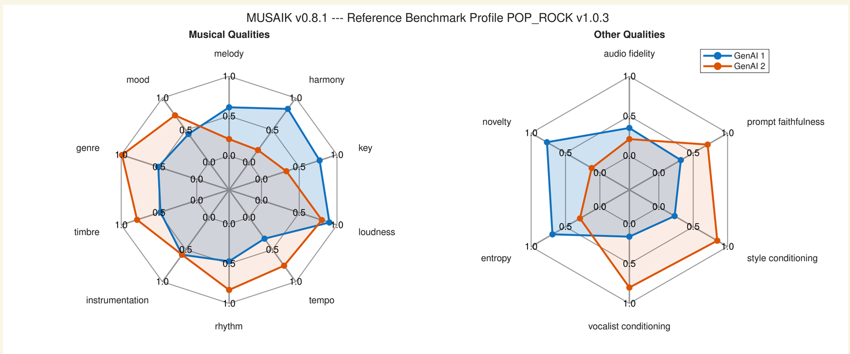

Figure: Mockup of aggregated result presentation.

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- **Logging:**
  - auditable logs (machine & human readable formats) with RBP, version info, test data, etc.

Limitations

- Missing **subjective evaluation protocols**
- **Black-box evaluation**:
  - no evaluation of the underlying model architecture/capabilities
  - no distinction between human contributions and fully machine-generated output
- **Focus on output qualities**:
  - no usability or user experience evaluation
  - no evaluation other characteristics, such as explainability, bias, resource usage, ethical use of data, etc.

Cross-Domain Applicability

- easily transferable to other domains of generative AI, such as **Visual Art**, **Text Generation**, and **Game Content Generation**
- modular architecture ⇒ **exchange feature representations**.

References

Alexander Lerch, Claire Arthur, Nick Bryan-Kinns, Corey Ford, Qianyi Sun, and Ashvala Vinay.
Survey on the Evaluation of Generative Models in Music.
ACM Computing Surveys, 58(4):99:1–99:36, October 2025.